

Supporting America's AI Action Plan

The Bionic Enterprise: Architecting the Intelligent Society of the Future

*"The enterprise of the future won't **build** killer apps,
It will **become** the killer app."*

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AI-Driven Framework for the Future

The Bionic Enterprise: Architecting the Intelligent Society of the Future and its companion *Transformation Leader Guide* by the same title provide a detailed implementation roadmap and reference architecture for envisioning and implementing AI-driven organizations. This framework and Bionic Transformation approach should be integrated as a central component the *United States AI Action Plan* to help the United States achieve and maintain strategic and tactical advantage across industry, academia, national defense, and the intelligent society. The framework addresses the concept of USA 2.0 (Bionic Cities, Bionic Transportation, Bionic National Defense, Bionic Healthcare, Bionic Homes, Bionic Companies, and more) as a comparable initiative to Japan's Society 5.0 efforts. It describes the emergence of organizations that are supported by their own Bionic Digital Platforms populated with AI-driven Bionic Agents that animate specific Bionic Technologies that "fuse" Bionic capabilities with human processes and workflows at the most intimate levels of human activity. A Bionic Enterprise is ubiquitous, persistent, and encompasses a broad array of multi-model user interfaces and experiences. Augmented reality, virtual reality, synthetic reality are commonplace. The human-centric Bionic Enterprise is hyper-connected, hyper-aware, hyper intelligent, hyper-converged, and hyper-adaptive. Improving human and organizational performance and potential at all levels across all aspects of our society is its purpose.

A 2019 Gartner Study stated that “Agent interfaces represent a whole new paradigm of human-computer interaction and have broad implications that will greatly influence how enterprises interact with customers, offer services and provide tools to employees.”

There are many brilliant AI researchers and implementers across all industries and facets of society, but what is lacking is a comprehensive framework that addresses all aspects of an organization (human, technology, and transformation). This framework provides a means of blending the technologies and products from these innovative researchers and product developers into a holistic enterprise framework. The Bionic Enterprise Framework is not focused on any specific, current AI algorithm or technique, but on the *capabilities* that AI and the technologies of the Fourth Industrial Revolution bring. Tools and algorithms change at a blinding pace, but the anchor for designing the intelligent enterprise of the future is the set of capability maturity models that are used to define an organization’s roadmap for continuous transformation.

The Bionic Enterprise Technology Capability Maturity Model addresses functional capabilities that will eventually go beyond the current vision of artificial intelligence into artificial general intelligence, to synthetic intelligence, and eventually to the point of enterprise consciousness.

“The Bionic Enterprise is hyper-connected, hyper-aware, hyper-converged, and hyper-intelligent, and hyper-adaptive.”

The Bionic Enterprise Framework is not a notional concept that espouses vague musings about how AI will make everything better – or everything worse. It is a detailed and directly actionable reference model and transformation process for architecting and implementing the Bionic Enterprise of the future in any domain or field of human endeavor. All the homework is there, and yet like any effective framework, its adoption can be tailored to support any implementation and show exactly where and how every technology fits into the enterprise schema.

The framework focuses on the three types of crucial capital for the 21st century, *Cognitive Capital* (what organizations and their people know how to do), *Behavioral Capital* (what customers, suppliers, partners, or adversaries do), *Network Capital* (where and how users

of a Bionic Enterprise locate, interface, and interact with the Bionic Enterprise through a variety of multi-modal interfaces and experiences.

While the possibilities are limitless, we must still prepare for and anticipate the advent of *Dark AI* – a future sentient and synthetic intelligence that may arise as a digital mutation or intentional creation. Dark AI will not have humanity's best interest in mind. It may see us as an irrelevant obstacle to its own aspirations. Dark AI may well lead to a series of AI wars – digital and physical battles of good vs. evil with humanity caught in the middle. This is already happening in cyberspace and is spilling over onto the battlefields of the 21st century.

From Theory to Implementation

The Bionic Enterprise Framework is beginning to be adopted by graduate students across DoD in various organizations. They are learning to implement the concepts and thought process to enable system development, acquisition and operational activities. It is being used to define and architect the role and placement of Bionic Technologies and Bionic Capabilities (from the Bionic Capability Maturity Model) within an organization to define its Bionic Digital Platform. The two books and frameworks they describe are being used as textbooks for systems engineering at the Naval Postgraduate School for the System Engineering SE0495 capstone course and at Virginia Tech.

The book's foreword is by John Zachman, CEO of Zachman International and the creator of the Zachman Framework for Enterprise Architecture (the underlying framework for the Department of Defense Architecture Framework (DoDAF) and every other enterprise architecture framework). Dr. Nikhil Deshpande –Intel Corporation, Senior Director for AI and High-Performance Computing and Graph Analytics commented on the framework that it is *"...a framework for today's enterprises to leap into the bionic future in a responsible way."* Mike Walker –Microsoft Senior Director: Healthcare Lifesciences Digital Strategy said, *"The concept of a Bionic Enterprise is a critical concept to prepare organizations for the digital future that awaits."*

The Bionic Enterprise <http://www.BionicEnterprise.tech> is a framework of frameworks that address the three fundamental perspectives of any organization: The Human Perspective, the Technology Perspective, and the Fusion Perspective, where technology and human activity meet to form a Bionic Enterprise. The URL above provides many large downloadable "canvases" that outline the concepts and sample use-cases of how the framework can be applied to begin to specify Bionic Functions, Bionic Technologies, Bionic Effects, and Bionic Capabilities to achieve those effects.

The concept of a Bionic Fusion Strand is introduced to define specifically how Bionic Functions enable Bionic Effects within a user experience to achieve improved outcomes and performance. Bionic Fusion Strand designs for different user experiences are the heart of the Fusion Perspective where technology meets human activity. Bionic Fusion Strands are the DNA of a Bionic Enterprise.

Human Perspective

Human Perspective of an organization begins with a Visionary Leadership Maturity Model, an organizational Creativity Maturity Model, and an overall Organizational Maturity Model. The focus on the Human Perspective helps organizations understand the importance and process for re-tooling and re-imagining their organizations as engines of continuous transformation. Without continuous transformation to high value-producing entities, organizations in any field or industry rapidly become irrelevant with the ever-shrinking business, technology and military cycle times for innovation and transformation. The Bionic Enterprise must also cultivate organizational and individual maturity to be comfortable with this change and embrace it. Without visionary leaders and visionary leadership to cultivate creativity and innovation in every corner of the enterprise, an organization cannot achieve full Bionic Transformation and strategic advantage. The alternative is individual silos of excellence with stove-piped solutions that do not fit purposefully into a holistic, integrated, Bionic Enterprise Digital Platform and transformation vision.

Technology Perspective

The Technology Perspective of the framework provides fully developed reference architecture frameworks required to architect the fusion of Bionic Technologies with human activity. The Technology Perspective begins with Bloom's Taxonomy – the intersection of Knowledge and Cognitive Processes used by educators to develop learning plans for students from elementary through postgraduate. The Bionic Enterprise Framework adopts Bloom's Taxonomy as a mechanism to identify the types of cognitive and computational capabilities to be realized in AI-driven Bionic Agents. These capabilities are described in a detailed Bionic Technology Capability Maturity model that spans five Bionic Capability Areas:

- Bionic Capability Area 1: Interface
- Bionic Capability Area 2: Intelligence and Cognition
- Bionic Capability Area 3: Integration
- Bionic Capability Area 4: Platform and Infrastructure
- Bionic Capability Area 5: Transformation & Architecture

The Bionic Technology Capability Maturity Model is an element of the Technology Perspective of the framework and spans these capability areas. The cognitive capabilities described progress in maturity and culminate with AI eventually reaching the point of *The Singularity* – the point at which computers evolve to a point of synthetic intelligence that becomes as intelligent or more intelligent than humans. The framework addresses the implications of the Singularity and strategies for coping with such an outcome. Until that point of AI evolution is reached, however, organizations can use the Bionic Enterprise Framework to describe, define, architect, and transform to achieve and maintain relevance and even dominance over peers, or adversaries.

“The Bionic Enterprise has opened my eyes to a new reality of transformation: one that fuses the human and technological perspectives.” - Warren Vaneman, PhD, Professor – ESSP, Adjunct Professor Virginia Tech

Also within the Technology Perspective is a taxonomy of Bionic Technologies, a taxonomy of Bionic Agent types, and most crucially, a taxonomy of detailed Bionic Functions. The Bionic Functions of an agent define how the action verbs in Bloom’s Taxonomy will be implemented and embodied within individual AI-driven Bionic Agents to achieve the Bionic Effects required to improve different user experiences across the enterprise.

Fusion Perspective: Transformation and Architecture

Automating and digitalizing processes and activities isn’t good enough. The Bionic Enterprise focuses on value effects (Bionic Effects) and outcomes that can be achieved within enterprise processes or user experiences at enterprise scale. We must design the intelligent society of the future as community of cooperating Bionic Enterprises so that technology interfaces and interaction experiences enable desirable outcomes. The transformation to a Bionic Enterprise is more than just Digital Transformation. Bionic transformation results in integrated Bionic Digital Platforms with converged data, hyper-connectivity, and distributed intelligence at the edge, and those capabilities are accessible through the previously mentioned multi-modal interfaces and interactions. The Bionic Fusion Framework provides a comprehensive, integrated mechanism and process for conducting on-going enterprise-wide assessments using the different maturity models of the Bionic Enterprise Framework. Assessments begin with the leadership, culture, organization and value streams and value-generation mechanisms of the organization. These targeted value streams and user experiences are a starting point for identifying opportunities for user experience transformation using the Bionic User Experience Design

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Process. The result is a series of detailed Bionic Fusion Strands that describe the fusion of the Bionic Agents with human activities. The Fusion Perspective also provides the Bionic Enterprise Transformation Lifecycle to drive continuous assessment and transformation. Think of it as Enterprise-scale Bionic DEVOPS.

The resultant Bionic Enterprise must exude the core characteristics: Interactive & Immersive, Intelligent, Adaptive, Ubiquitous, Goal Directed, Elastic.

America's Bridge to the AI-Driven Bionic Future

The paradigm shift to this intelligent future requires organizations to detect or even cause paradigm shifts to achieve strategic advantage within their eco-systems. The Bionic Enterprise Framework informs the development of a Bionic Strategy and Roadmap, implementation of visionary leadership, the creative and innovative enterprise, and the purposeful implementation of disruptive technologies. The fusion of human activity and Bionic Technologies will transform society, education, finance, industry, and the future of work and wealth. Bionic Transformation requires organizational and cultural change and paradigm shifts to envision, embrace, and architect the future. The Bionic-inspired future features advanced, immersive, and empathetic human interfaces to Bionic Agents and infrastructure that blur the line between cyberspace and the physical world. New hyper-intelligent, human-centric companies and societies will evolve with the Bionic Enterprise Framework as their blueprint.

The United States AI Strategy must embrace this comprehensive approach. It has already been recognized and praised by leading industry figures in related fields of digital transformation, robotics, system engineering, and enterprise architecture. Maintaining our position as the leader of the free world and keeping our adversaries at bay without a comprehensive framework for rapidly implementing continuous transformation is like taking on a colossal engineering feat without a blueprint or a plan to implement it. The Bionic Enterprise provides the vision, strategy, and framework for USA 2.0.

A deeper dive into the Bionic Enterprise can be found in a Defense Acquisition University (DAU) webinar I conducted in 2024 titled: The Bionic Enterprise – Framework for the Future

<https://www.dau.edu/events/bionic-enterprise-framework-future>

Kevin Brett Bio

Kevin Brett is a Certified Enterprise Architect (CEA Blackbelt) through the FEAC Institute, a Scaled Agile Framework (SAFe) Architect. He is an IT professional with over forty years in

government, defense and commercial IT project and portfolio management, space shuttle mission experience, and enterprise architecture. He has developed and taught a comprehensive, practicum-based, Black Belt Enterprise Architecture Certification course.

Kevin has managed and led enterprise architecture programs, digital acquisition and engineering eco-system development, software, and systems engineering efforts, process automation and re-engineering for forty years. He has developed software and system engineering and enterprise architecture methodologies and training programs. His background includes development of AI rule-based systems in the fields of medical diagnosis for the U.S. Air Force. He developed an AI air traffic flow management adviser system, Smartflo, for the FAA utilizing an architecture of cooperating rule-based software agents to make recommendations for traffic flow management.

Kevin also led the development of a space shuttle data communications rule-based AI system. The system diagnosed communications down link issues from the Spacelab module during space shuttle missions. Kevin has developed object-oriented modeling and agent-based architecture methodologies and agent-based systems with embedded AI-agents. He spent the first half of his career as computer programmer, software engineer and mission analyst on a variety of projects including supporting numerous space shuttle missions as a member of the Space Shuttle Spacelab Team. He also developed communications and scheduling software for the NASA Communications (NASCOM) Network Control Center (NCC).

Kevin implemented a software engineering methodology he developed, Gecko Lite, at NASA Ames Research Center on the NASA Grid Computing program to lend systems and software engineering rigor to the development efforts of the program. In support of this effort, he developed an extensive program of instruction to train contractors and NASA engineers and developers on all aspects and phases of software and systems engineering.

Kevin has directly implemented and led development efforts for implementation of process re-engineering efforts using process automation tools to automate more than a dozen internal corporate processes and several customer business processes. These re-engineering efforts resulted in dramatic reductions in cycle-times and increased throughput and for process requests through automated tracking and notification of process events.

Kevin has led enterprise architecture teams for organizations in private industry and government including the Federal Aviation Administration (ATO), Environmental Protection Agency, the United States Marine Corps Enterprise Architecture, and the United States Marine Corps Enterprise Network Architecture. His focus and interest are on creative

strategies for organizational and digital transformation supported by enterprise architecture and emerging technologies.

Kevin is concerned especially with strategic planning based on understanding the current and future desired states of aspects of enterprise maturity as it pertains to digital transformation and construction of digital eco-systems. He has a keen interest in helping executives, architects, and engineers develop the tools to understand and overcome transformation challenges and how paradigm shifts in operations, technology, culture, and marketplace can affect organizational direction, structure, and operations. He has developed analytical methods to help enterprises detect, understand, or even initiate their own paradigm shifts to achieve and maintain leadership positions within their domain.

Kevin holds a B.S. in Computer and Information Science from the University of Maryland, University College.